

# LAB 5

## Interfacing LCD with I2C

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### Objectives:

To interface a 16x2 LCD with the TIVA C Series microcontroller (TM4C123G) and display text messages.

### Material:

- TIVA C Series Launchpad
- LCD 1602
- 10k $\Omega$  potentiometer (for LCD contrast adjustment)
- Bread board
- Wires

### Submission Instructions:

- Demonstrate your work for each activity to your instructor. Missing the instructor signature results in **20% penalty**.
- Submit a zipped file of your project folder onto the BB dropbox. Missing the zipped project file results in a **20% penalty**.
- Write a lab report detailing the implementation and results. Include the schematic and picture of your circuit, screenshot of your code and results. Answer all the questions.

### Theory:

I2C is a synchronous serial communication protocol using SDA (data) and SCL (clock). The Tiva C acts as master and LCD (via PCF8574) acts as slave. Typical addresses are 0x27 or 0x3F.

### Pin Configuration:

- LCD VCC  $\rightarrow$  5V
- LCD GND  $\rightarrow$  GND
- LCD SDA  $\rightarrow$  PB3
- LCD SCL  $\rightarrow$  PB2

### Lab Activity 1 [20 points]

1. Connect the I2C LCD module to the **Tiva C LaunchPad** according to the given circuit configuration.
2. Adjust the **potentiometer** to obtain proper LCD contrast.
3. Compile and upload the provided source code file, **LCD\_I2C\_Template**, to the microcontroller.

The file is available on **Blackboard** under the **Additional Resources** module and in the **Lab Dropbox**.

4. Verify that the LCD displays the message: “**Hello Tiva**”.
5. Modify the program so that the same text is displayed at **line 2, column 5** of the LCD.

### Lab Activity 2 [20 points]

1. Modify the previous program so that the text “**Counter:**” is displayed **centered on the first line** of the LCD.
2. Display an **incrementing counter value** centered on the **second line**.
3. Update the counter value at an interval of **1 second**.

### Lab Activity 3[30points]

1. Modify both the **software** and the **hardware connections** to use **I2C1** instead of **I2C0** for LCD interfacing.
2. Display a message of your choice on the LCD using the new I2C module.
3. In your lab report, clearly explain all modifications made, including:
  - changes in pin connections
  - changes in peripheral initialization
  - changes in GPIO configuration
  - changes in the I2C base/module used in the program
  - any additional code changes required for successful operation

## Question 1 [30 points]

1. Why are pull-up resistors required in I2C instead of pull-down resistors?
2. What would happen if SDA is held LOW by a faulty device on the bus?
3. Compare I2C with SPI in terms of:
  - a. hardware complexity
  - b. speed
  - c. scalability
4. How does the master detect whether a slave device is present on the bus?
5. What is the address of I2C LCD?
6. Can we connect multiple I2C devices (e.g., sensor + LCD) on the same bus?
7. If two devices have the same I2C address, how can this conflict be resolved?
8. How does the LCD differentiate between command and data?
9. Why is a delay required during LCD initialization?
10. What changes are required to port this code from Tiva C to another MCU (e.g., STM32)?